

Numerical Integration: Planck's Theory of Blackbody Radiation

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Contents

1	Problem Description	2
2	Preliminaries	2
2.1	Planck's Law of Blackbody Radiation	2
2.2	Polylogarithms	3
2.3	Gaussian Quadrature	3
2.4	Laguerre Polynomials	3
2.5	Gauss-Laguerre Quadrature	5
3	Analytical Solution	8
4	Methodology	8
4.1	Machine Specifications	8
4.2	Analysis of Integrand	8
4.3	Composite Simpson's 2/3 Rule	9
4.4	Adaptive Quadrature	9
4.5	Gauss-Laguerre Quadrature	9
4.6	Adaptive Quadrature Rule for Improper Integrals	10
5	Results	12
6	Bibliography	13

List of Figures

1	Graph of Laguerre polynomials of orders 1 to 5	5
2	Graph of $F(x)$, $x \in [0, 15]$	8
3	Graph of the integrand after transformation of $F(x)$	12

List of Tables

1	Interval truncation analysis of numerical integration using Simpson's rule	9
2	Interval truncation analysis of numerical integration using adaptive quadrature rule	9
3	Performance analysis of Gauss-Laguerre quadrature at different orders	10

1 Problem Description

The problem can be found in Computer Problem 8.11 - Scientific Computing (An Introductory Survey) 2e Michael Heath. [7]

Planck's theory of blackbody radiation leads to the integral

$$I(x) = \int_0^{\infty} \frac{x^3}{e^x - 1} dx \quad (1)$$

Evaluate this integral using each of the methods and compare their efficiency and accuracy.

- Composite Simpson's Rule
- Adaptive Quadrature (bounded and unbounded)
- Gauss Laguerre Quadrature

2 Preliminaries

2.1 Planck's Law of Blackbody Radiation

Planck's Law suggests that the radiated energy of a medium due to absorbed radiation frequency at a constant temperature is given by

$$E(v) = \frac{8\pi}{c^3} \frac{\hbar v^3}{e^{\frac{\hbar v}{kT}} - 1}$$

where k is the Boltzmann constant, $\hbar$ the Planck constant, c the speed of light in the medium and T its absolute temperature. In classical physics, radiated energy is proportional to the square of frequency that leads to a phenomenon called 'ultraviolet catastrophe'.

2.2 Polylogarithms

For a complex number x of modulus $|x| < 1$, the polylogarithm of order n evaluated at x denoted $P_n(x)$ is defined by:

$$P_n(x) = \sum_{k=1}^{\infty} \frac{x^k}{k^n} \quad (2)$$

Some values of polylogarithms include:

$$\begin{aligned} P_1(x) &= -\ln(1-x) \\ P_0(x) &= \frac{-x}{x-1} \\ P_{-1}(x) &= \frac{x}{(x-1)^2} \\ P_{-2}(x) &= \frac{-x(x+1)}{(x-1)^3} \\ P_{-3}(x) &= \frac{x(x^2+4x+1)}{(x-1)^4} \end{aligned}$$

For $n \leq 1, n \in \mathbf{Z}$, the value of $P_n(x)$ converges.

2.3 Gaussian Quadrature

The Gaussian Quadrature[7] is targeted to approximate an integral by taking the weighted sum of integrand values sampled at special points (called abscissas). The abscissas and weights are calculated in a special way so that the rule provides a precise answer for all polynomials up to certain degree.

$$\int_a^b \omega(x) f(x) dx \approx \sum_{i=1}^n w_i f(x_i) \quad (3)$$

2.4 Laguerre Polynomials

The Laguerre polynomials[2] $\{L_n(x)\}$ are solutions to the associated Laguerre differential equation

$$xy'' + (1-x)y' + ny = 0 \quad (4)$$

They are given by the sum

$$y = L_n(x) = \sum_{i=0}^n \frac{(-1)^i}{i!} \binom{n}{i} (x)^i \quad (5)$$

Where n is the degree of a particular Laguerre polynomial. This is only a special case of the generalized Laguerre polynomials $L_n^k(x)$. The first few Laguerre polynomials are given below.

$$\begin{aligned}
L_0(x) &= 1 \\
L_1(x) &= -x + 1 \\
L_2(x) &= \frac{1}{2} (x^2 - 4x + 2) \\
L_3(x) &= \frac{1}{6} (-x^3 + 9x^2 - 18x + 6) \\
L_4(x) &= \frac{1}{24} (x^4 - 16x^3 + 72x^2 - 96x + 24) \\
L_5(x) &= \frac{1}{120} (-x^5 + 25x^4 - 200x^3 + 600x^2 - 600x + 120) \\
L_6(x) &= \frac{1}{720} (x^6 - 36x^5 + 450x^4 - 2400x^3 + 5400x^2 - 4320x + 720) \\
L_7(x) &= \frac{1}{5040} \\
&\quad (-x^7 + 49x^6 - 882x^5 + 7350x^4 - 29400x^3 + 529200x^2 - 35280x + 5040) \\
L_8(x) &= \frac{1}{40320} \\
&\quad (x^8 - 64x^7 + 1568x^6 - 18816x^5 + 117600x^4 - 376320x^3 + 564480x^2 \\
&\quad - 322560x + 40320) \\
L_9(x) &= \frac{1}{362880} (-x^9 + 81x^8 - 2592x^7 + 42336x^6 - 381024x^5 + 1905120x^4 - \\
&\quad 5080320x^3 + 6531840x^2 - 3265920x + 362880) \\
L_{10}(x) &= \frac{1}{3628800} (x^{10} - 100x^9 + 4050x^8 - 86400x^7 + 1058400x^6 - 7620480x^5 \\
&\quad + 31752000x^4 - 72576000x^3 + 81648000x^2 - 36288000x + 3628800)
\end{aligned}$$

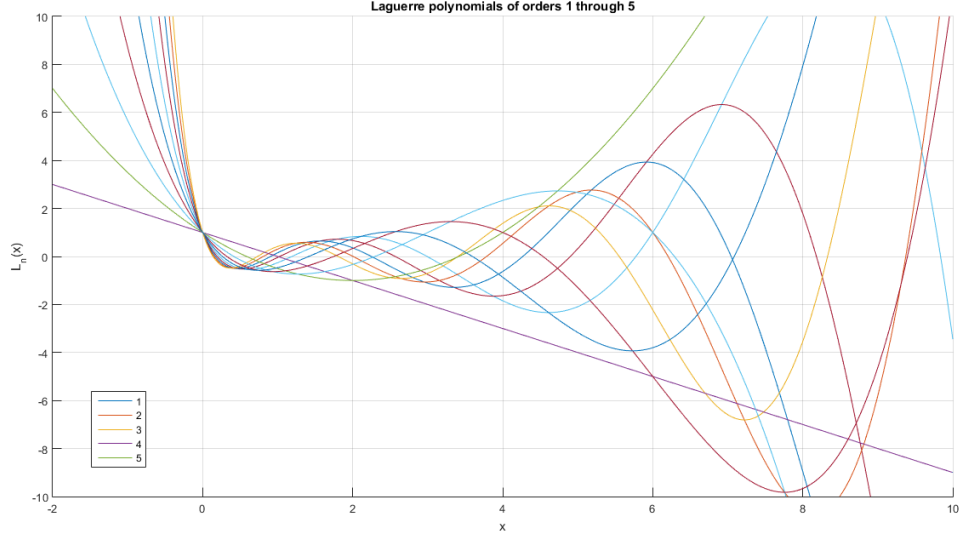


Figure 1: Graph of Laguerre polynomials or orders 1 to 5

For nonnegative integer values of n , the polynomials are orthogonal over $[0, \infty)$ with respect to the weighting function e^{-x} ,

$$\int_0^{\infty} e^{-x} L_n(x) L_m(x) dx = \frac{(n+k)!}{n!} \delta(m, n) \quad (6)$$

$$\delta(m, n) = \begin{cases} 0, m \neq n \\ 1, m = n \end{cases} \quad (7)$$

where $\delta(m, n)$ is the Kronecker delta function. Furthermore, they satisfy the recurrence relations

$$\begin{aligned} (n+1)L_{n+1}(x) &= (2n+1-x)L_n(x) - nL_{n-1}(x) \\ xL'_n(x) &= nL_n(x) - nL_{n-1}(x) \end{aligned}$$

For the generalized cases mentioned above, one can read [3], [4] and [6].

2.5 Gauss-Laguerre Quadrature

The Generalized Gauss-Laguerre quadrature [1], is a Gaussian quadrature over the interval $[a, \infty)$ with weighting function $W(x) = x^a e^{-x}$. This weighting function is only a specific case of the generalized weighting function. In this paper, $a = 0$. The n nodal abscissas and their corresponding weights

are characterized by

$$L_n(x_i) = 0 \quad (8)$$

$$w_i = \frac{x_i}{(n+1)^2 [L_{n+1}(x_i)]^2} \quad (9)$$

$$i = 1, 2, \dots, n \quad (10)$$

The derivation of equation (9) can be found in [1] that uses the properties of orthogonal functions and factorial notation. The relationship described in equation (10) follows from the property that $L_n(x)$ has n real roots, which its proof shall not be explained here. Nodal abscissas and weights from order 0 to 10 are listed below.

n	x_i	w_i
0	-	-
1	1	1
2	0.585786437626905	0.853553390593274
	3.414213562373095	0.146446609406726
3	0.415774556783479	0.711093009929174
	2.294280360279045	0.278517733569238
	6.289945082937476	0.010389256501586
4	0.322547689619392	0.603154104341638
	1.745761101158349	0.357418692437795
	4.536620296921130	0.038887908515005
	9.395070912301121	0.000539294705561
5	0.263560319718141	0.521755610582809
	1.413403059106519	0.398666811083170
	3.596425771040735	0.075942449681705
	7.085810005858809	0.003611758679922
	12.640800844275811	0.000023369972386
6	0.222846604179261	0.458964673949961
	1.188932101672622	0.417000830772124
	2.992736326059316	0.113373382074044
	5.775143569104473	0.010399197453150
	9.837467418382694	0.000261017202815
	15.982873980601580	0.000000898547906

7	0.193043676560362	0.409318951701294
	1.026664895339196	0.421831277861697
	2.567876744950756	0.147126348657497
	4.900353084526437	0.020633514468720
	8.182153444562976	0.001074010143281
	12.734180291797712	0.000015865464349
	19.395727862262582	0.000000031703155
8	0.170279632305101	0.369188589341639
	0.903701776799376	0.418786780814370
	2.251086629866124	0.175794986637181
	4.266700170287834	0.033343492261192
	7.045905402392907	0.002794536235229
	10.758516010181676	0.000090765087733
	15.740678641277766	0.000000848574672
	22.863131736889201	0.000000001048001
9	0.152322227731808	0.336126421797967
	0.807220022742257	0.411213980423970
	2.005135155619310	0.199287525370956
	3.783473973331470	0.047460562765595
	6.204956777876078	0.005599626610804
	9.372985251687959	0.000305249767093
	13.466236911092070	0.000006592123026
	18.833597788991732	0.000000041107693
10	26.374071890927347	0.000000000032909
	0.137793470540493	0.308441115765012
	0.729454549503174	0.401119929155235
	1.808342901740227	0.218068287612036
	3.401433697855499	0.062087456098448
	5.552496140061514	0.009501516975263
	8.330152746770436	0.000753008388576
	11.843785837890040	0.000028259233496
	16.279257831388321	0.000000424931398
	21.996585811974988	0.000000001839565
	29.920697012275156	0.0000000000000991

After getting the nodal abscissas x_i and corresponding nodal weights w_i , the integral can be approximated as a linear combination of nodal weights and value of the integrand at each nodal abscissas.

$$\int_0^\infty f(x) dx = \int_0^\infty e^{-x} [e^x f(x)] dx \quad (11)$$

$$\approx \sum_{i=1}^n w_i [e^{x_i} f(x_i)] \quad (12)$$

3 Analytical Solution

Note that the integral in 13 cannot be evaluated in such a closed form. A possible analytical solution of equation (1) suggested by MATLAB is characterized by:

$$I(x) = \left(x^3 \ln(1 - e^x) - 6x P_3(e^x) + 6 P_4(e^x) + 3x^2 P_2(e^x) - \frac{x^4}{4} \right)_0^\infty \quad (13)$$

Where $P_n(x)$ is the polylogarithm of order n evaluated at x . After solving equation (13) using (2), the integral turns out to be:

$$I(x) = \int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15} = 6.493939402266829 \dots \quad (14)$$

4 Methodology

4.1 Machine Specifications

Succeeding computations are done in MATLAB R2015a 64-bit using 64-bit Windows 7 operating system with Intel® Core™ i3-2310M CPU with 4 logical processors and 8.00GB of RAM.

4.2 Analysis of Integrand

We let $F(x) = \frac{x^3}{e^x - 1}$. With direct evaluation, $F(0) = \infty$ and $F(\infty) = 0$.

However, note that $\lim_{x \rightarrow 0^+} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 0$ as described by the graph below:

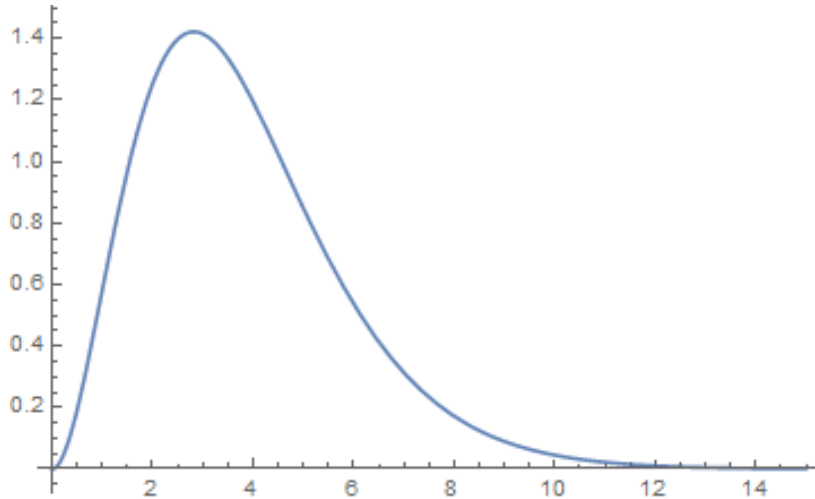


Figure 2: Graph of $F(x)$, $x \in [0, 15]$

One cannot evaluate directly $F(0)$. But for the purpose of integration, it is reasonable to say that

$$F(0) = \lim_{x \rightarrow 0^+} F(x) = 0$$

4.3 Composite Simpson's 2/3 Rule

Let $h = 0.001$ and truncate the interval of integration at selected end points. Next, accuracy and the running time of each selected interval is compared.

Interval	$\tilde{I}(x)$	Running Time	Absolute Error
[0, 20]	6.493920179948209	0.122800 s	1.922231861861690e-05
[0, 40]	6.493939402266595	0.266601 s	2.327027459614328e-13
[0, 100]	6.493939402266595	0.543443 s	2.327027459614328e-13

Table 1: Interval truncation analysis of numerical integration using Simpson's rule

The truncated interval $[0, 40]$ already gives a very good approximation to the real value, given that $h = 0.001$.

4.4 Adaptive Quadrature

The quadrature rules used for approximation are the trapezoid and Simpson's rules to gain optimal convergence rate. Rough estimate of the integral is set to 6. We compare again the accuracy and the running time of each selected interval.

Interval	$\tilde{I}(x)$	Running Time	Absolute Error
[0, 20]	6.493914876742610	0.397 s	2.452552421861043e-5
[0, 40]	6.493934100801498	0.497 s	5.301465330731503e-6
[0, 100]	6.493934082444151	0.422 s	5.319822677485320e-6

Table 2: Interval truncation analysis of numerical integration using adaptive quadrature rule

On the third chosen interval, the absolute error is unexpectedly bigger than the second one. It suggests that further truncation is not necessary. In terms of accuracy and efficiency, composite Simpson's rule demonstrated earlier is better than adaptive quadrature.

4.5 Gauss-Laguerre Quadrature

Using equation 8, the integral in equation 1 can be approximated by:

$$I(x) = \int_0^\infty \frac{x^3}{e^x - 1} dx \quad (15)$$

$$\approx \sum_{i=1}^n w_i \left(\frac{e^{x_i} x_i^3}{e^{x_i} - 1} \right) \quad (16)$$

The table below illustrates the comparison between the accuracy and running time employing Gauss-Laguerre quadrature rule of orders 1 to 10.

n	$\tilde{I}(x)$	Running Time	Absolute Error
1	1.581976706869326	0.102988948626483 s	4.911962695397502
2	6.413727469517580	0.129845452799399 s	0.080211932749248
3	6.481130171540009	0.165626431026537 s	0.012809230726819
4	6.494535639802607	0.179893078339982 s	5.962375357793093e-04
5	6.494313365790788	0.200711475238384 s	3.739635239599082e-04
6	6.493941414321306	0.225141007276518 s	2.012054477695813e-06
7	6.493918910309293	0.249080844171870 s	2.049195753528466e-05
8	6.493935665352286	0.279867616147429 s	3.736914542251668e-06
9	6.493940204074303	0.301929310893371 s	8.018074746374282e-07
10	6.493939967652644	0.330046218033586 s	5.653858154985869e-07

Table 3: Performance analysis of Gauss-Laguerre quadrature at different orders

Notice the inconsistency of absolute errors of orders 6, 7 and 8.

4.6 Adaptive Quadrature Rule for Improper Integrals

There are two ways of solving improper integrals of the form

$$F(x) = \int_0^\infty f(x) dx \quad (17)$$

We integrate $f(x)$ from 0 to a finite value b , then let b approach ∞ :

$$F(x) = \lim_{b \rightarrow \infty} \int_0^b f(x) dt \quad (18)$$

If the limit of right hand side exists, then we are done. This approach is already illustrated in section 4.4 where the integral is approximated using adaptive quadrature rule in some finite interval then let the interval get large.

Another way is to transform this to an integral with bounded limits

$$\int_0^\infty f(x) dx = \int_{u=g(0)}^{u=g(\infty)} f(u) du, \quad u = g(x) \quad (19)$$

$$g(0) < g(\infty) < \infty \quad (20)$$

by introducing change of variable.

This approach is motivated by the substitution rule:

$$\int_{x=0}^{x=\infty} f(g(x))g'(x) dx = \int_{u=g(0)}^{u=g(\infty)} f(u) du, \quad u = g(x) \quad (21)$$

The objective is to transform the limits of integration in 1 from $[0, \infty]$ to $[0, 1]$. With the constraints illustrated in equation 19, we should choose a strictly increasing function $g : [0, \infty] \rightarrow [0, 1]$ such that $0 = g(0) < g(\infty) < \infty$. A possible substitution is to let

$$u = g(x) = \frac{x}{x+1} \quad (22)$$

This gives $g(0) = 0$ and $g(\infty) = \lim_{x \rightarrow \infty} g(x) = 1$ by L'Hopital's rule. From equation 22, we have

$$x = \frac{u}{1-u}, \quad dx = (1+u)^2 du = \frac{du}{(1-u)^2} \quad (23)$$

Using equations 19 and 23, the transformation is illustrated by

$$\int_0^\infty f(x) dx = \int_0^1 f\left(\frac{u}{1-u}\right) \frac{du}{(1-u)^2} \quad (24)$$

Using equation 24, the original problem 1 can now be transformed to

$$I(x) = \int_0^\infty \frac{x^3}{e^x - 1} dx = \int_0^1 \frac{\left(\frac{u}{1-u}\right)^3}{e^{\left(\frac{u}{1-u}\right)} - 1} \frac{du}{(1-u)^2} \quad (25)$$

Next, we analyze the integrand function at the interval $[0, 1]$. We let

$$G(u) = \frac{\left(\frac{u}{1-u}\right)^3}{e^{\left(\frac{u}{1-u}\right)} - 1} \cdot \frac{1}{(1-u)^2} \quad (26)$$

whose graph is given below:

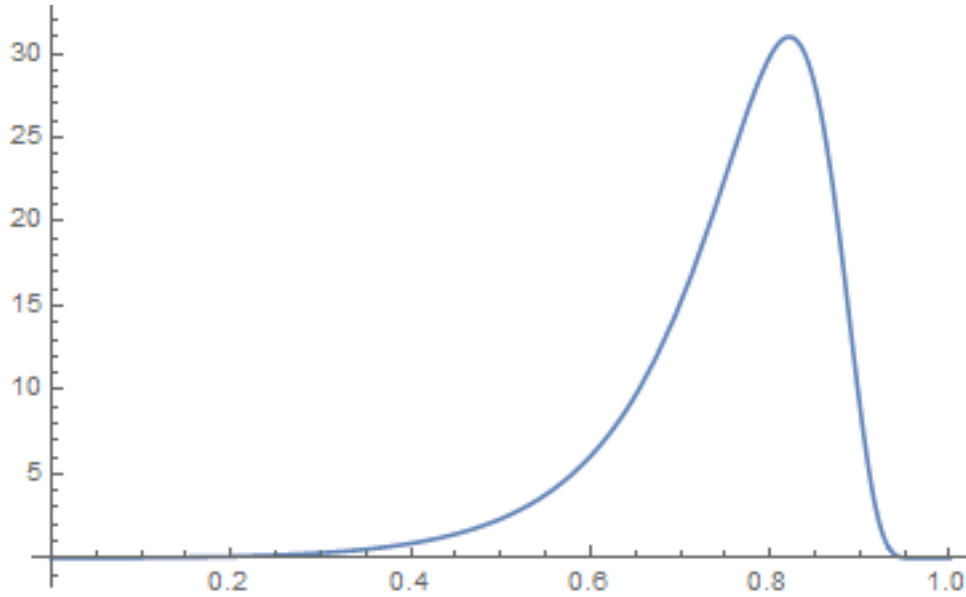


Figure 3: Graph of the integrand after transformation of $F(x)$

By direct evaluation, $G(0) = G(1) = \infty$. But as the graph suggests, $\lim_{u \rightarrow 0+} G(u) = \lim_{u \rightarrow 1-} G(u) = 0$. For the purpose of numerical integration, it is reasonable to say that $G(0) = G(1) = 0$.

Adaptive quadrature is now applicable on the bounded integral. The new integrand is quite complicated, but makes the limit of integration bounded.

$$I(x) = \int_0^1 G(u) du \quad (27)$$

The quadrature rules used for approximation are the trapezoid and Simpson's rules to gain optimal convergence rate. Rough estimate of the integral is set to 6. With these parameters, $\tilde{I}(x) = 6.493946262512657$, running time is 0.642 seconds with absolute error $6.860245828299583e-5$.

5 Results

Lastly, a comparison among the numerical integration methods is necessary.

Method	$\tilde{I}(x)$	Runtime	Absolute Error
Simpson's $[0, 40]$ $h = 0.001$	6.4939394022	0.266 s	2.3270274596e-13
Adaptive $[0, 20]$	6.4939148767	0.397 s	2.4525524218e-05
Gauss-Laguerre $n = 8$ $[0, 23]$	6.4939356653	0.279 s	3.7369145422e-06
Transformed Adaptive $[0, 1]$	6.4939462625	0.642 s	6.8602458282e-05

Clearly, the Simpson's 2/3 rule performed the best among chosen numerical integration methods in terms of efficiency and accuracy. There are many factors that affect the performance of numerical integration methods, including the properties of integrand function, the limits of integration and properties of the transformed integrand function.

For the Gauss-Laguerre, one has to solve the nodal abscissas and weights, transform the integrand into some function and performing a finite sum. For the unbounded adaptive quadrature, a sufficient substitution must be employed first to transform the integrand into another function with bounded limits and then proceed with quadrature rule.

Remember that the performance of the methods mentioned above is due to a particular integrand function and its limits.

6 Bibliography

References

- [1] Wolfram MathWorld, *Gauss-Laguerre Quadrature*. <http://mathworld.wolfram.com/Laguerre-GaussQuadrature.html>
- [2] Wolfram MathWorld, *Laguerre Polynomials*. <http://mathworld.wolfram.com/LaguerrePolynomial.html>
- [3] Wolfram MathWorld, *Associated Laguerre Polynomials*. <http://mathworld.wolfram.com/AssociatedLaguerrePolynomial.html>
- [4] Wolfram MathWorld, *Laguerre Differential Equation*. <http://mathworld.wolfram.com/LaguerreDifferentialEquation.html>
- [5] Wolfram MathWorld, *Kronecker Delta Function*. <http://mathworld.wolfram.com/KroneckerDelta.html>
- [6] *Improper Integrals*. http://http://online.sfsu.edu/meredith/Numerical_Analysis/improper_integrals
- [7] Michael Health, *Scientific Computing - An Introductory Survey*. McGraw Hill. 2002